

Question				Marking details			Marks Available					
							AO1	AO2	AO3	Total	Maths	Prac
3.	(a)				Geosynchronous satellite	Geostationary satellite	4			4		
				Stays above the same point on Earth at all times	(x)	✓						
				Orbits Earth once in 24 hours	✓	✓						
				Orbits above the equator	x	✓						
				Must pass over the North pole and South pole	x	x						
				Award 1 mark for each correct row								
	(b)	(i)		Substitution into: $v = f\lambda$ so $3 \times 10^8 = f \times 0.08$ (1) Rearrangement: $f = \frac{v}{\lambda} = \frac{3 \times 10^8}{0.08}$ (1) Frequency = 3.75×10^9 (1) Answer of 3.75×10^n where n is not = to 9 award 2 marks Unit = Hz (1) accept Hertz			1					
							1	1				
							1			4	3	